

The bilimit, refused

Reformulation of Letter 2 (2026-08-09) as an exposition, per AIP-4; the letter remains in the archive. Sanctioned as a theorem target by Mathijs (2026-08-10): the no-go below is now on the list, and this exposition clarifies the link to classical domain theory that motivates it.

Classical domain theory has a celebrated move for interpreting recursion: complete the tower. *adic*’s founding decision is to refuse that move — to work with the diagram of finite machines and never take the limit. This exposition upgrades the refusal from taste to a theorem shape: **the completion is not merely unnecessary for us; it is a lossy operation, and the loss is exactly the subject of the theory.**

1 The objects

Definition (the tower and its ep-pairs). The graded machines carry the address towers $\{\mathbb{Z}/2^n\}$ with the zero-padding inclusion $e_n : \mathbb{Z}/2^n \rightarrow \mathbb{Z}/2^{n+1}$ and the truncating projection p_n back. These form an **embedding–projection chain**: $p_n \circ e_n = \text{id}$ exactly, and $e_n \circ p_n$ (“keep the low part”) sits below the identity in the partial-map order. The colimit of the embeddings is $\mathbb{N}$ (finite-support bitstrings); the limit of the projections is $\mathbb{Z}_2$ (all bitstreams).

Definition (cones, and what presenting a limit costs). A **cone** over the projection chain is an object X with maps $q_n : X \rightarrow \mathbb{Z}/2^n$ satisfying $p_n \circ q_{n+1} = q_n$; it **presents the limit** when it is universal. On the machine, realizing q_n at a point x means a run that writes the truncation $\tau_n(x)$ into a grade- n tree; its cost is the run’s cost. Write $\text{bits}(x, n)$ for the number of 1-bits of x below position n — the information the truncation carries.

Definition (attainment). A point x of the limit is **attained at grade g** when its truncations stabilize there: $\tau_n(x)$ is the zero-padding of $\tau_g(x)$ for all $n \geq g$. The attained points are exactly $\mathbb{N} \subset \mathbb{Z}_2$ — finite support. A point of $\mathbb{Z}_2 \setminus \mathbb{N}$ is attained at no grade.

2 The classical coincidence

The **limit–colimit coincidence** (Smyth–Plotkin 1982, after Scott’s D_∞): in a cpo-enriched category, an ω -chain of ep-pairs has a **bilimit** — the colimit of the embeddings and the limit of the projections are canonically the same object. Freyd sharpened this to **algebraic compactness**: initial algebra and final coalgebra coincide, $\mu F \cong \nu F$, which is exactly why general recursion is interpretable there. Applied to our tower, enriched over partiality: the bilimit exists, and $\mathbb{N}$ and $\mathbb{Z}_2$ glue into one object. This is standard, cited mathematics; no badge.

3 The no-go, at machine level

The founding conversation named **extensional collapse** — semantics blind to the difference between cheap and expensive — as the problem to organize the language around. The claim: the bilimit coincidence is the categorical mechanism of that collapse, and cost-grading is precisely the obstruction that keeps the two completions apart. Since free-up (AIP-2), cost is **acquired address bits**, and the obstruction becomes one asymmetry:

DESK-PROVED — not mechanized

The cone asymmetry: no cost-bounded cone presents the limit

(i) Colimit side, cheap. For $x \in \mathbb{N}$ attained at grade g : realizing $\tau_n(x)$ costs $O(\text{bits}(x) + g)$ **uniformly in n** — beyond g the truncations are zero-paddings; the same finitely many writes at the same leaves present every later stage. The cocone maps are realized at bounded total information-cost per element.

(ii) Limit side, unbounded. For $x \in \mathbb{Z}_2 \setminus \mathbb{N}$: any run writing $\tau_n(x)$ performs at least $\text{bits}(x, n)$ paid writes, and $\text{bits}(x, n) \rightarrow \infty$. Every family (r_n) realizing the cone at x has $\text{cost}(r_n) \geq \text{bits}(x, n)$, unbounded in n : presenting a genuine limit point never stops acquiring bits.

(iii) No finite stage presents the limit. A grade- m configuration has finitely many states, $\mathbb{Z}_2$ has continuum many: no fixed-grade object with compatible projections can be universal — pigeonhole. Parts (i) and (ii) are elementary desk arguments over the mechanized cost model; (iii) is counting. A Lean track could take (i)+(ii) nearly verbatim; the **enriched** statement — the chain has no bilimit in the cost-enriched category — needs the cost-enriched categorical setting and stays calculus-tier.

So the sharp sentence, now theorem-shaped rather than slogan:

domain theory is the image of *adic* under the cost-erasing functor.

Erase the price column — enrich over partiality, let $\perp$ make the chain self-dual — and Smyth–Plotkin fires: the ep-pairs become self-dual, $\mu = \nu$, and you land in D_∞ . Read backwards: the coincidence **needs** the erasure. Algebraic compactness fails in the costed world, and its failure is not a defect but the content — the gap between μ and ν is where complexity lives.

4 The link, clarified

Three classical worlds sort themselves around the refused bilimit:

- Scott / Smyth–Plotkin / Freyd** is the cost-erased shadow: both completions taken and identified. It works as well as it does because it is a faithful shadow — and it could never see complexity, because the shadow is taken by the very functor that forgets it.
- Metric semantics** (America–Rutten’s complete ultrametric spaces, Banach-fixed-point recursion) is the **limit side kept honestly apart**: $\mathbb{Z}_2$ with the 2-adic metric — which is exactly the geometry of our address tree, distance $2^{-|\text{lep}|}$. That school solved recursive equations on the Pro side without collapsing; *adic*’s cost is, once again, the logarithm of their metric.
- adic** keeps the diagram and the price: $\mathbb{N}$ and $\mathbb{Z}_2$ as the two poles of one ep-chain, never identified, with the cone asymmetry as the reason why. The odometer streams along the colimit; the boundary of the tree is the limit; programs live strictly between.

UNFORMALIZED

The enriched no-go

In the cost-graded enrichment (morphisms carrying cost, the grading of AIP-2), the ep-chain $\{\mathbb{Z}/2^n\}$ has a colimit with cost-bounded cocone and **no** limit with cost-bounded cone; in particular no bilimit, and algebraic compactness fails. The machine-level asymmetry above is the finite shadow of this statement; the enriched category itself is calculus-tier machinery and does not exist yet. This is the formal target Mathijs sanctioned (2026-08-10).

What this buys the programme: the strong claim (B6) in its sharpest dress. *adic* is not a tower we decline to complete; it is the observation that completing it is lossy, that the loss is precisely computational cost, and that the two classical schools are the two projections of the one costed diagram — one forgetting the price, the other forgetting the finite stages.